

11 (**short**) A rubber ball of mass m bounces off a hard horizontal floor, as shown in Fig. 8. The ball makes contact with the floor at time $t = 0$ with a downward speed U and it leaves the floor at time $t = t_1$ with an upward speed V . This contact period is shown shaded in the graph of the motion. During the bounce the contact with the floor is modelled as a spring k in parallel with a dashpot λ .

For this question you should refer to the impulse response information on page 7 of the Mechanics Data Book. You may assume that the damping ratio $\zeta \ll 1$.

(a) Show that $t_1 \approx \frac{\pi}{\omega_n}$ where $\omega_n = \sqrt{\frac{k}{m}}$. [3]

(b) The coefficient of restitution for the bounce is defined as $e = \frac{V}{U}$, as shown in the graph of the motion.

Show that $e \approx 1 - a\zeta$ and find the value of the constant a . [7]

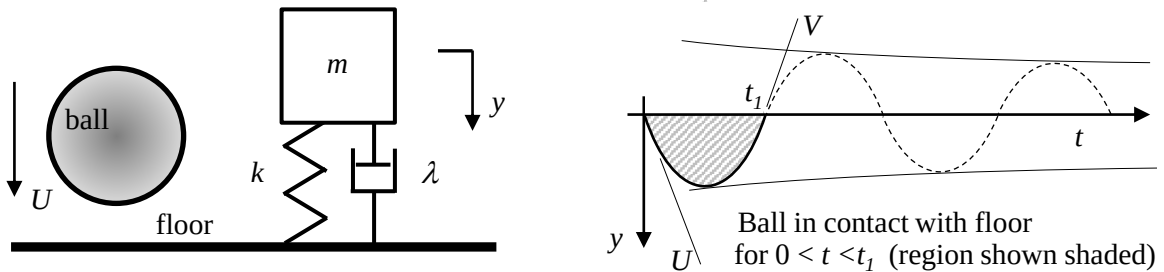


Fig. 8